STS2 BI pipeline: end-to-end audit

Re-audit: 12-13 Oct 2026 (first audit 11 Oct) | Scope: local save files to Power BI report | Branch: main (PR #1 merged 12 Oct)

Summary

The pipeline is correct end to end and now complete: all 1,061 runs on disk reconcile through GCS, the raw BigQuery tables, the dbt marts and the Power BI model, and the report's headline numbers match independent BigQuery queries exactly. 167 dbt checks run on every build (166 pass, 1 expected warning), the daily build is now live on main, and every finding rated High or Medium in the first audit is closed. This audit found one significant gap and fixed it: the game saves modded play in a separate folder that the uploader never read, so 115 runs (11%, including every run since 30 Aug) were missing. Loading that backlog exposed a related weakness, also fixed: with a single loader instance and no retry policy, bursts of uploads were silently dropped. What remains is low-risk tidying plus one scalability item: each run now takes the loader about a minute.

Status by stage

Stage	Status	Verdict
1. Extraction (local saves + uploader)	**GREEN**	Vanilla and modded save trees read; 1,061 files, 0 pending
2. Scheduling (Task Scheduler)	**GREEN**	Monthly with catch-up on; next run 1 Nov; needs the PC on
3. Landing (GCS)	**GREEN**	1,061 objects (115 modded), 62.2 MB
4. Transform + load (Cloud Function)	**AMBER**	Correct, least-privilege, alerting, retries; ~1 min per run limits backlog speed
5. Raw warehouse (BigQuery sts2)	**GREEN**	1,061 runs, no duplicate keys, partitioned + clustered
6. Modelling + tests (dbt)	**GREEN**	167 checks: 166 pass, 1 expected warning; contracts on all marts
7. CI / access (GitHub Actions, IAM)	**AMBER**	Daily build live and keyless; 1p budget alert; default compute SA still holds Editor
8. Semantic model + report (Power BI)	**GREEN**	Reads sts2_marts (+ local art only); star schema; numbers reconcile
9. Repo hygiene + docs	**GREEN**	No secrets, raw data, cache or art files tracked (screenshots: F20); docs current

Priority actions

When	Actions
Done	F15 modded saves loaded and tagged; F16 retry policy; F10 1p budget alert; F1-F3, F5, F6, F9, F13 from the first audit
Next	F17 batch the loader (or ELT); F18 dev target for local dbt; F4 remove Editor from the default SA
Later	F7 catalogues into dbt; F11 scope dbt-runner; F8 all players' characters

Reconciliation chain

Check	Result	Source
Local .run files (profile1, vanilla + modded)	**946 + 115 = 1,061**	File count; profile2 excluded by decision
Uploader manifest entries	**1,061**	scripts/uploaded_manifest.json; 0 pending
GCS objects (raw/, raw/modded/)	**1,061**	62.2 MB total
Raw sts2.fact_run rows / distinct run_id	**1,061 / 1,061**	BigQuery
Mart dim_run rows (Vanilla / Modded)	**1,061 (946 / 115)**	BigQuery
Power BI Total Runs (no slicers)	**1,061**	Power BI engine (DAX)

Check	Result	Source
Win rate: BigQuery vs report	**15.9% / 15.9%**	169 wins / 1,061 runs
Win rate (completed): BigQuery vs report	**20.1% / 20.1%**	169 / (1,061 - 221 abandoned)
Avg floors reached: BigQuery vs report	**23.4 / 23.4**	dim_run.floors_reached
Avg death floor: BigQuery vs report	**20.8 / 20.8**	dim_run.death_floor
Event choices: BigQuery vs report	**9,541 / 9,541**	fct_event_choice
Potion events: BigQuery vs report	**26,317 / 26,317**	fct_potion_event
Fights: BigQuery vs report	**18,940 / 18,940**	fct_combat
Rest-site choices: BigQuery vs report	**6,606 / 6,606**	fct_rest_choice

Child facts legitimately cover fewer runs than dim_run (1,061): floor states 1,055 (runs abandoned within seconds have no floors - tested and excluded), fights 1,034, card choices 1,030, event choices 1,045. Date range 2026-03-05 to 2026-10-12.

Strengths

• Full reconciliation: 1,061 runs at every stage from disk to dashboard; four headline KPIs and four fact-table row counts match independent BigQuery queries exactly.

• 167 dbt checks on every build: strict source tests, referential integrity from every fact to its dimensions, grain-uniqueness tests, enforced contracts on every mart, and domain checks.

• Layered warehouse: raw sts2 (loader-owned) -> sts2_staging (views) -> sts2_marts (tested tables). Power BI reads only sts2_marts for data; game artwork is a local presentation asset, never committed.

• Clean star schema around dim_run: every fact relates to dimensions, never to another fact.

• Second data source done properly: reference data (cards, events, relics, potions, encounters, Ancient dialogue) is read from the installed game files, replacing a wiki scrape whose terms forbade it.

• Operability: fail-fast uploader, least-privilege loader account, log-based alerting, retry policy, and an idempotent loader that makes any reprocessing safe.

• Keyless CI: GitHub Actions uses Workload Identity Federation restricted to this repository; the daily dbt build now runs from main.

• Cost: well inside free tiers (MB-scale GCS and BigQuery, free GitHub Actions on a public repo).

Findings register

Closed since the first audit: F1 (bucket variable set, uploader fails fast), F2 (missed runs caught up), F3 (log-based alerting; plus F16 retries), F5 (PR #1 merged, daily build live), F6 (300 s timeout), F9 (manifest saved per file), F13 (docs corrected), and the wiki-scrape terms issue (reference data now read from the game's own files).

F15 HIGH Extraction [Resolved]

The game writes modded play to a separate save tree (steam/<id>/modded/profile1/saves/history) that the uploader never read: 115 runs from May to 12 Oct were missing, including every run since 30 Aug.

Evidence	Newest run in the vanilla folder 30 Aug; 115 files in the modded folder, none in the manifest or GCS.
Recommendation	Done: the uploader reads profile1 from both trees, runs are tagged save_source (Vanilla / Modded) from their GCS path and exposed in dim_run; profile2 (10 old test runs) excluded by the owner's choice.
Effort	Done

F16 MEDIUM Cloud Function [Resolved]

With max-instances=1 and RETRY_POLICY_DO_NOT_RETRY, deliveries arriving while the instance was busy were aborted ('no available instance') and the runs silently dropped.

Evidence	Cloud Run warnings on 11 Oct and again during the backlog upload.
Recommendation	Done: trigger redeployed with --retry; safe because the loader is idempotent per run_id.
Effort	Done

F17 MEDIUM Cloud Function [Open]

Each run costs the loader ~28 sequential BigQuery jobs (delete + load for 13 tables, 2 catalogue MERGEs), about a minute per run with one instance, so the 115-run backlog took around two hours to drain. Fine for monthly volumes, slow for backfills.

Evidence	Observed load rate during the backlog: about one run per minute.
Recommendation	Batch per run (one multi-statement transaction, or load all tables then one DELETE per table), or move flattening into dbt (true ELT) so the function only lands files. For one-off backfills, scripts/backfill_table.py already loads a whole table in one job.
Effort	2-4 h

F18 LOW dbt / CI [Open]

Local dbt builds write to the same datasets as CI. A local build running at the same moment as a PR workflow briefly replaced a seed table, and the CI run failed ('relic_catalog was not found').

Evidence	PR run 38317686459 failed at 22:38 while a local build ran; the run on main a minute later passed.
Recommendation	Add a dev target writing to sts2_dev_* datasets for local work; keep the shared datasets for CI.
Effort	30 min

F19 LOW Cloud Function [Resolved]

One 4-player modded run made a 'Key Door' event choice on a boss floor, where the room records only the fight, so the choice had no event_id. The strict not_null source test stopped the build.

Evidence	source_not_null_sts2_fact_event_choice_event_id: FAIL 4 (one run, four players).
Recommendation	Done: flatten.py falls back to the event named in the choice key; table reloaded, loader redeployed, all tests pass. A good example of a strict source test catching new data.
Effort	Done

F4 LOW IAM [Partly resolved]

The loader now runs as a dedicated least-privilege account, but the default compute service account still holds roles/editor (plus BigQuery and Storage roles) although nothing uses it.

Evidence	Project IAM policy.
Recommendation	Remove Editor and the unused grants from 448620880316-compute@... .
Effort	5 min

F7 LOW Cloud Function [Open]

Card and relic catalogues are still MERGEd by the loader; this raced under concurrency once and is why max-instances must stay at 1 (and part of F17).

Evidence	cloud_function/main.py _upsert_dimensions.
Recommendation	Derive the catalogues in dbt from the facts and drop the MERGE; concurrency can then rise.
Effort	1-2 h

F8 LOW Cloud Function [Partly resolved]

Deprecated utcfromtimestamp fixed; fact_run.character still only captures players[0].

Evidence	cloud_function/flatten.py.
Recommendation	Model all players' characters (fct_run_player) when co-op analysis matters.
Effort	30 min

F10 LOW Cost / ops [Resolved]

There was no budget alert for the project.

Evidence	Budget 'STS2 BI - any spend (1p)': GBP 0.01/month on sts2-bi-1995ebj, alerts at 100% of actual and forecast spend (gcloud stored 0.01GBP as 1 nano-pound; corrected via the API and verified).
Recommendation	Done 13 Oct: any spend at all emails the billing admins. Budgets alert, they don't cap.
Effort	Done

F11 LOW IAM [Open]

dbt-runner holds bigquery.dataEditor at project level rather than on the datasets it uses.

Evidence	Project IAM policy.
Recommendation	Scope to sts2 (read) and sts2_staging / sts2_marts (write).
Effort	15 min

F20 INFO Repo [Open]

Seeds include short game text from Mega Crit's localization (event, option, potion and encounter text, Ancient dialogue). Artwork files are extracted locally and never committed, but the report PDF and screenshots in docs/ show the artwork as page backgrounds.

Evidence	dbt/seeds/*.csv; powerbi/assets/ is gitignored; docs/report.pdf and docs/screenshots/*.jpg.
Recommendation	Acceptable for a non-commercial portfolio; if asked, regenerate the text columns locally and export the report without background art (or drop the screenshots).
Effort	-

F12 INFO Power BI [Accepted]

Refresh is manual in Desktop; scheduled refresh needs a Pro licence.

Evidence	No published dataset.
Recommendation	Accepted for a personal project.
Effort	-

F14 INFO dbt [Accepted]

3 cards appear in runs but not in the current game files (Grapple, Follow Through, Scare - retired). Reported by the warn-level coverage test, as designed.

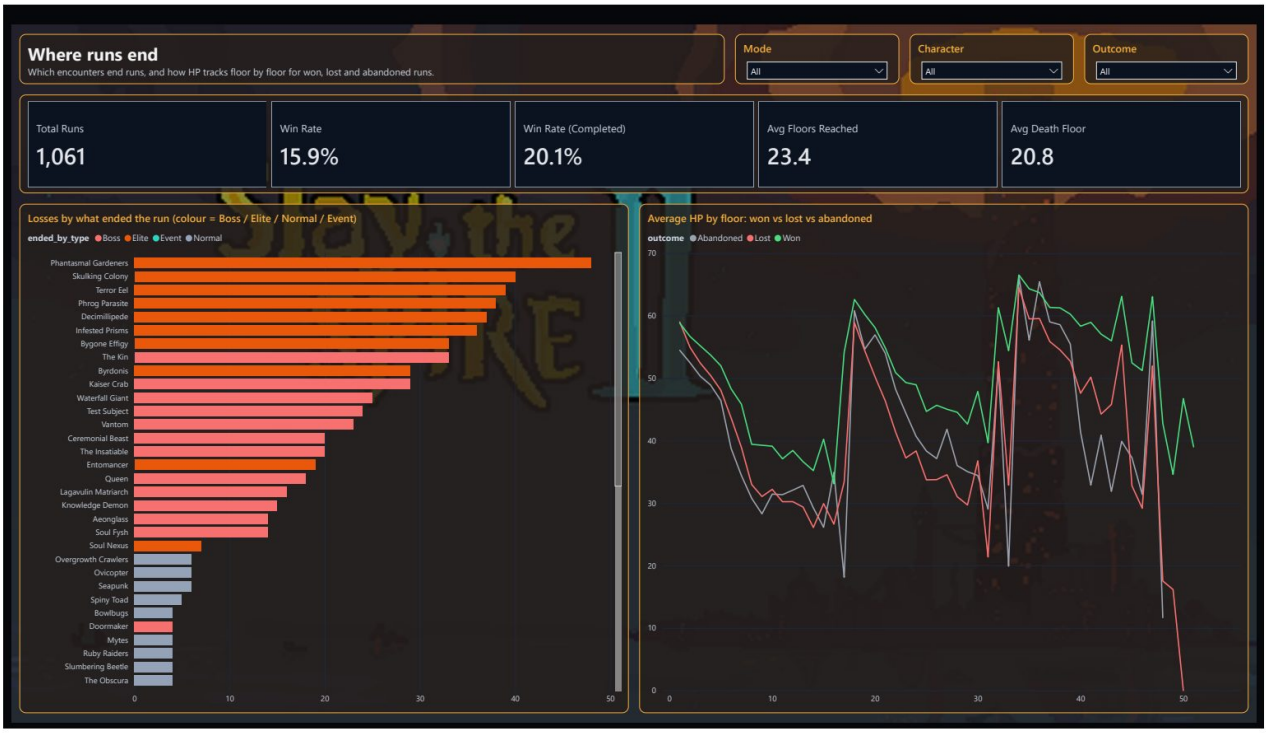
Evidence	assert_cards_in_catalog: WARN 3.
Recommendation	No action; re-run scripts/build_reference_seeds.py after game patches.
Effort	-

Method

Each stage was checked against live systems: local save folders and Task Scheduler (PowerShell, schtasks), GCS contents and Cloud Function configuration and logs (gcloud), IAM policy and alert policies, row counts and KPI queries in BigQuery, a full dbt build, GitHub Actions run history, and the Power BI model queried directly through its local Analysis Services engine, plus the rendered report.

Appendix: report pages

Where runs end



Fights & rest sites



Solo vs Multiplayer



Card picks



Events



Potions



Explore


Explore
Every run, one row each. Sort or filter any column.
Search encounters
All
Mode
All
Character
All
Outcome
All
Every run, newest first - rows tinted by character; outcome in green / red / grey
Date Character Mode Party Ascension Outcome Floors Act reached Ended by Type Run time (min) Save
12/10/2026 Defect Multiplayer 2 players 10 Lost 46 3 Aeonglass Boss 74.50 Modded
12/10/2026 Defect Solo Solo 9 Lost 17 1 Waterfall Giant Boss 4.70 Modded
12/10/2026 Ironclad Multiplayer 2 players 10 Lost 31 2 The Insatiable Boss 40.00 Modded
11/10/2026 Defect Solo Solo 5 Lost 17 1 The Kin Boss 6.10 Modded
11/10/2026 Defect Solo Solo 10 Lost 13 1 Skulking Colony Elite 6.40 Modded
11/10/2026 Defect Solo Solo 10 Lost 17 1 The Kin Boss 24.90 Modded
11/10/2026 Ironclad Multiplayer 2 players 10 Abandoned 1 1 0.10 Modded
11/10/2026 Ironclad Multiplayer 2 players 10 Lost 27 2 Decimillipede Elite 37.80 Modded
11/10/2026 Ironclad Multiplayer 3 players 10 Won 46 3 75.40 Modded
11/10/2026 Ironclad Solo Solo 10 Lost 17 1 Waterfall Giant Boss 12.50 Modded
11/10/2026 Ironclad Solo Solo 10 Lost 33 2 Kaiser Crab Boss 26.60 Modded
11/10/2026 Ironclad Solo Solo 10 Won 49 3 41.90 Modded
11/10/2026 Necrobinder Multiplayer 2 players 10 Lost 14 1 Phrog Parasite Elite 19.80 Modded
11/10/2026 Necrobinder Solo Solo 10 Lost 12 1 Terror Eel Elite 10.10 Modded
10/10/2026 Defect Solo Solo 10 Lost 8 1 Terror Eel Elite 6.70 Modded
10/10/2026 Necrobinder Solo Solo 10 Lost 11 1 Skulking Colony Elite 7.30 Modded
10/10/2026 Silent Solo Solo 5 Abandoned 35 3 20.40 Modded
10/10/2026 Silent Solo Solo 10 Lost 17 1 The Kin Boss 15.70 Modded
09/10/2026 Ironclad Multiplayer 2 players 10 Lost 33 2 Kaiser Crab Boss 37.50 Modded
09/10/2026 Ironclad Solo Solo 10 Lost 33 2 The Insatiable Boss 40.90 Modded
08/10/2026 Defect Solo Solo 10 Abandoned 24 2 21.70 Modded
08/10/2026 Ironclad Solo Solo 1 Lost 48 3 Test Subject Boss 19.70 Modded
08/10/2026 Silent Solo Solo 10 Lost 23 2 Spiny Toad Normal 20.60 Modded
07/10/2026 Ironclad Multiplayer 4 players 10 Lost 46 3 Aeonglass Boss 81.60 Modded
07/10/2026 Ironclad Solo Solo 0 Abandoned 32 2 15.20 Modded
07/10/2026 Ironclad Solo Solo 10 Lost 15 1 Skulking Colony Elite 11.70 Modded
06/10/2026 Necrobinder Solo Solo 9 Abandoned 6 1 3.10 Modded
06/10/2026 Necrobinder Solo Solo 9 Abandoned 16 1 9.20 Modded
06/10/2026 Necrobinder Solo Solo 9 Abandoned 36 3 18.20 Modded
06/10/2026 Necrobinder Solo Solo 9 Lost 15 1 Skulking Colony Elite 4.90 Modded
06/10/2026 Silent Solo Solo 10 Lost 24 2 Entomancer Elite 21.70 Modded
05/10/2026 Necrobinder Solo Solo 7 Lost 11 1 Phantasmal Gardeners Elite 5.30 Modded
Total
Outcome mix
Won 15.93%
Abandoned 20.83%
Lost 63.24%
Runs by character
Ironclad 305
Defect 213
Silent 192
Regent 187
Necrobinder 164
0 50 100 150 200 250 300 350